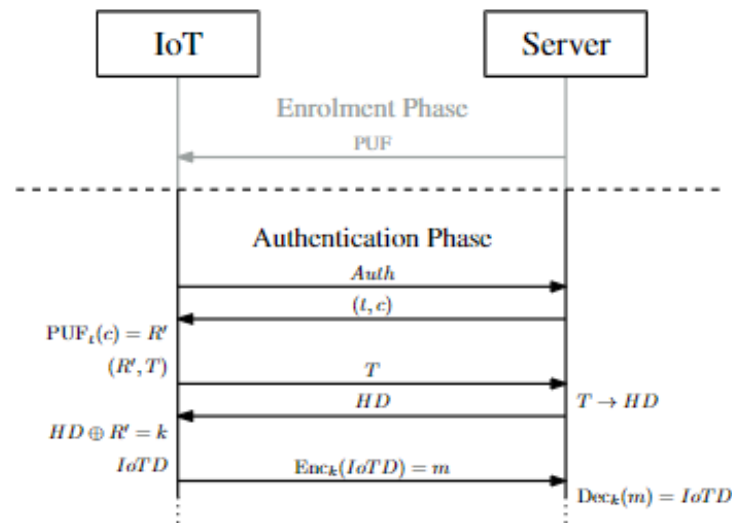




- A Physical Unclonable Function based on the retention characteristics of a Dynamic Random-Access Memory is utilised.
- During the enrollment phase, the characteristics of the PUF are recorded by the server. These include the challenges (c) and the responses (R) for particular decay times (t) and temperatures (T), as well as associated helper data (HD) and keys (k).
- Then, the PUF is provided to a particular IoT device/entity.
- The enrollment phase takes place only once in a secure environment.
- In the authentication phase, a response (R') is produced by the PUF (at the device/entity's side) for a particular challenge (c) and a specific decay time (t) selected by the server. The relevant temperature (T) is reported to the server and the corresponding helper data (HD) are sent to the IoT device/entity to reproduce a key (k) and use it to encrypt IoT measurement data.
- The relevant data are decrypted at the server's side using the same key (k); the server can produce this key (k) as it knows the helper data (HD) as well as the challenge (c), decay time (t) and temperature (T) that led to the production of the relevant PUF response at the device's side (R').
- The authentication phase can be repeated as many times as needed.